Environmental Studies (EVS)

QUIZ 1 FOR FA1

1. Which pollutant reduces crop productivity primarily by interfering with the opening and closing of stomata, leading to reduced CO2 intake?

A. SO2
B. O3
C. NO2
D. CO
Ans. B. O3

2. In a grassland ecosystem, if 10,000 kJ of energy is available at the producer level, approximately how much energy would be available to tertiary consumers?

A. 100 kJ
B. 10 kJ
C. 1,000 kJ
D. 1 kJ
Ans. B. 10 kJ

3. Which statement is true regarding ecological pyramids?

A. Pyramid of numbers is always upright
B. Pyramid of energy can never be inverted
C. Pyramid of biomass is always upright in aquatic ecosystems
D. Pyramid of energy may be inverted in forests
Ans. B. Pyramid of energy can never be inverted

4. In the food chain (Phytoplankton > Zooplankton > Small Fish > Large Fish > Birds), which has the highest concentration of DDT?

A. Zooplankton
B. Phytoplankton
C. Large Fish
D. Bird
Ans. D. Bird

5. Which combination of pollutants causes photochemical smog?

A. SO2 and CO
B. NOx and hydrocarbons
C. O3 and CO2
D. Methane and CFC
Ans. B. NOx and hydrocarbons

6. Mangrove ecosystems provide several services. Which is incorrect?

A. Coastal protection, preventing erosion
B. Biodiversity—habitat for diverse flora and fauna
C. Climate regulation
D. Nutrient depletion—reduction of soil fertility
Ans. D. Nutrient depletion—reduction of soil fertility

7. Assertion–Reason: Assertion (A): NPP is always less than GPP. Reason (R): A part of GPP is used by plants for respiration.

A. Both A and R are true
B. Both A and R are false

C. A is true but R is false
D. A is false, R is true
Ans. A. Both A and R are true

8. A river near a town shows high BOD and fish deaths. Likely cause?

A. Algal bloom (eutrophication)
B. High dissolved oxygen from photosynthesis
C. Industrial effluents with biodegradable waste
D. Thermal pollution
Ans. C. Industrial effluents with biodegradable waste

9. Which ecosystem is most stable?

A. Desert
B. Arctic tundra
C. Tropical rainforest
D. Agricultural monoculture
Ans. C. Tropical rainforest

10. PM2.5 particles containing black carbon contribute to

A. Cooling of the atmosphere by reflecting sunlight
B. Warming of the atmosphere by absorbing radiation
C. Reduction of ozone depletion
D. Formation of acid rain only
Ans. B. Warming of the atmosphere by absorbing radiation

11. In Delhi during winter, PM2.5 concentration often exceeds 300 microgram/m^3. Which factor contributes most significantly to this seasonal spike?

A. Increased rainfall
B. Thermal inversion trapping pollutants
C. Higher wind
D. Formation of acid rain only
Ans. B. Thermal inversion trapping pollutants

12. What is the relation between NPP and GPP?

Ans. NPP = GPP - R

13. Identify two major indoor sources of PM2.5 used in rural households.

Ans. Fuel lamp and burning of wood (as entered in the submitted quiz)

14. What neurological disorders have been connected with PM2.5 pollution?

Ans. Alzheimer (as entered in the submitted quiz)

15. Propose two sustainable practices that help restore balance in biogeochemical cycles disturbed by human activity.

Ans. No answer supplied in the quiz (response: "NA")

16. "Energy flow is unidirectional while nutrient flow is cyclic." Explain.

Ans. The submitted response explains that energy flows through producers and consumers in one direction, while nutrients move cyclically through biotic and abiotic components.

17. National Air Quality Index (AQI) in India categorises parameters based on:

A. Only particulate concentration
B. Only gaseous pollutants
C. Particulates + gaseous pollutants + health impact

D. Meteorological parameters only

Ans. C. Particulates + gaseous pollutants + health impact

18. Primary challenge in urban rainwater harvesting is:

A. Low rainfall
B. Space and contamination issues
C. Excess storage capacity
D. Lack of rooftops

Ans. B. Space and contamination issues

19. How can cities use AI and IoT-based waste monitoring systems to improve waste collection efficiency?

Ans. No answer supplied in the quiz (response: "NA")